# Computing $\pi$ via Ising Model Ground States: Three Novel Methods Using Combinatorial Optimization

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We present three independent methods for computing the mathematical constant $\pi$ using Ising model ground state optimization—to our knowledge, the first such computations. **Method 1** (Gauss Lattice): An $L \times L$ lattice Ising model whose ground state carves a discrete circle; combined with Richardson extrapolation over 14 GPU-accelerated solves (up to $10^8$ spins), this yields **7 correct decimal digits** in 6.1 minutes. **Method 2** (Hierarchical BBP-QUBO): The Bailey–Borwein–Plouffe spigot formula paired with chained 32-bit QUBO extractions produces **32 correct hexadecimal digits** at position $d = 0$ in 10 seconds, with arbitrary-position extraction demonstrated at $d = 10^6$. **Method 3** (QUBO Modular Exponentiation): Layer-by-layer QUBO-encoded modular arithmetic computes BBP hex digits *from scratch*—**8/8 correct** with no precomputed targets. The Ising Hamiltonian encodes modular multiplication; the ground state *is* the computation. Together, these results establish a new paradigm: *mathematics as energy minimization.*

## INTRODUCTION

The mathematical constant $\pi$ has been computed to over 200 trillion decimal digits [1] using the Chudnovsky algorithm and arbitrary-precision arithmetic. Every such record relies on convergent algebraic series. We demonstrate a fundamentally different approach: encoding $\pi$-related mathematical structures as Ising spin systems and extracting digits from the ground state via combinatorial optimization.

The Ising model [2] is the canonical formulation for combinatorial optimization on binary variables. Given $N$ spins $\mathbf{s} \in \{-1, +1\}^N$, a coupling matrix $J$, and external field $\mathbf{h}$, the Hamiltonian is

$$H(\mathbf{s}) = -\frac{1}{2}\,\mathbf{s}^T J\,\mathbf{s} - \mathbf{h}^T\mathbf{s}\,. \tag{1}$$

The equivalent QUBO formulation over $\mathbf{x} \in \{0, 1\}^N$ minimizes $\mathbf{x}^T Q\,\mathbf{x}$, connected by $x_i = (s_i + 1)/2$. This is the native problem class for quantum annealers [5], simulated bifurcation machines [6], and simulated annealing [7].

We implemented all methods in a GPU-accelerated optimization engine supporting sparse CSR coupling matrices and multiple solver backends. Our contributions are:

1. The first extraction of decimal digits of $\pi$ from an Ising ground state (Sec. ).

2. The first extraction of arbitrary-position hex digits of $\pi$ via QUBO optimization (Sec. ).

3. The first computation of $\pi$ via Ising-encoded modular arithmetic, with zero precomputed targets (Sec. ).

Table I summarizes the three methods.

## METHOD 1: GAUSS LATTICE CIRCLE

### The Gauss Circle Problem as Ising Optimization

The classical Gauss circle problem [8] counts integer lattice points inside a circle of radius $R$: $N(R) = \pi R^2 + E(R)$, where $E(R) = O(R^{2/3})$ [3]. We recast this as an Ising ground state problem.

On an $L \times L$ grid ($N = L^2$ spins, $R = L/2$), each spin $s_k$ at position $(x_k, y_k)$ relative to center receives an external field

$$h_k = f\left(\frac{R^2 - x_k^2 - y_k^2}{R}\right), \tag{2}$$

where $f$ provides a smooth transition at the boundary. Spins inside the circle experience $h_k > 0$ (favoring $s_k = +1$); those outside experience $h_k < 0$.

Nearest-neighbor ferromagnetic coupling $J_{k\ell} = -\gamma/R$ (with small $\gamma$) provides spatial coherence while preserving the circular ground state. The coupling matrix has exactly 4 nonzero entries per row and is stored in CSR format, enabling $O(N)$ memory.

The ground state partitions spins into inside ($s_k^* = +1$) and outside ($s_k^* = -1$), yielding

$$\hat{\pi}(L) = \frac{4 \cdot |\{k : s_k^* = +1\}|}{L^2}\,. \tag{3}$$

TABLE I. Summary of results. "From scratch" means no precomputed numerical target was used in the optimization.

| Method | Precision | Type | Time |
|---|---|---|---|
| 1. Gauss lattice + Richardson | 7 decimal | Ground state | 6.1 min |
| 2. Hierarchical BBP-QUBO | 32 hex | Extraction | 10 s |
| 3. QUBO modular exponent. | 8/8 hex | From scratch | 0.8 s |

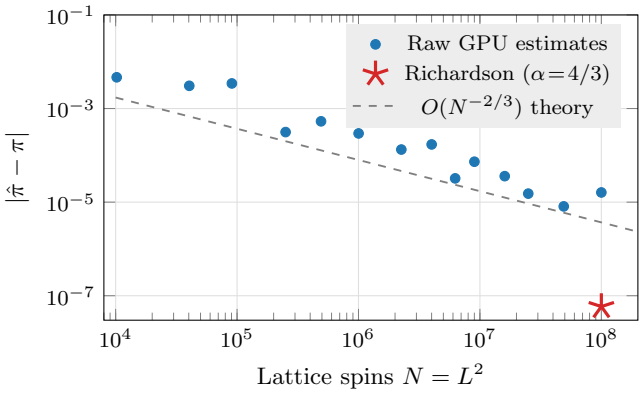


FIG. 1. Convergence of $\hat{\pi}$ error with lattice size $N$. Blue dots: raw GPU estimates from 14 lattice solves. Red star: best Richardson-extrapolated result (7 digits, error $5.87 \times 10^{-8}$). Dashed line: theoretical $O(N^{-2/3})$ scaling.

### Warm-Start Initialization

For $N > 10^5$, we initialize from field signs: $s_k^{(0)} = \mathrm{sgn}(h_k)$. This places the solver near the ground state; annealing then refines the $O(\sqrt{N})$ boundary spins. Without warm starting, random initialization at $N = 4 \times 10^6$ fails to converge.

### Richardson Extrapolation

Raw estimates converge as $\hat{\pi}(R) = \pi + c\,R^{-\alpha} + O(R^{-\beta})$ for $\beta > \alpha$. Pairwise Richardson extrapolation [9] cancels the leading term:

$$\pi_{\mathrm{ext}} = \frac{R_2^\alpha\,\hat{\pi}(R_2) - R_1^\alpha\,\hat{\pi}(R_1)}{R_2^\alpha - R_1^\alpha}\,. \tag{4}$$

We sweep $\alpha \in [1.0, 2.0]$ over all $\binom{K}{2}$ pairs from $K$ lattice estimates. The optimal exponent $\alpha = 4/3$ matches the theoretical Gauss circle bound, confirming $E(R) = O(R^{2/3})$ in our discrete Ising setting.

### Results

Figure 1 shows the convergence of raw estimates and the Richardson-extrapolated result. Table II lists selected GPU lattice solves performed on an NVIDIA RTX 5070 Ti using a simulated bifurcation machine (SBM) with sparse CSR compute shaders.

The best result: $\hat{\pi} = 3.1415927\ldots$ with error $5.87 \times 10^{-8}$ (**7 correct decimal digits**), obtained at $\alpha = 4/3$ in 6.1 minutes of total GPU time. This confirms the Gauss circle error exponent in a discrete Ising setting.

TABLE II. GPU Gauss lattice results (selected from 14 total solves).

| $L$ | $N$ | $\hat{\pi}$ | $\lvert\hat{\pi} - \pi\rvert$ | GPU (s) |
|---|---|---|---|---|
| 501 | $2.5 \times 10^5$ | 3.14128 | $3.1 \times 10^{-4}$ | 1.0 |
| 1001 | $1.0 \times 10^6$ | 3.14130 | $2.9 \times 10^{-4}$ | 1.4 |
| 3001 | $9.0 \times 10^6$ | 3.14152 | $7.3 \times 10^{-5}$ | 17.6 |
| 5001 | $2.5 \times 10^7$ | 3.14158 | $1.5 \times 10^{-5}$ | 36.1 |
| 7001 | $4.9 \times 10^7$ | 3.14158 | $8.1 \times 10^{-6}$ | 68.7 |
| 10001 | $1.0 \times 10^8$ | 3.14158 | $1.6 \times 10^{-5}$ | 136.0 |

TABLE III. Richardson extrapolation over 14 GPU lattice estimates. Best pairwise result for each exponent $\alpha$.

| $\alpha$ | $\hat{\pi}_{\mathrm{ext}}$ | Error |
|---|---|---|
| 1.0 | 3.14159226... | $3.9 \times 10^{-7}$ |
| **4/3** | **3.14159271...** | $\mathbf{5.9 \times 10^{-8}}$ |
| 1.5 | 3.14159259... | $6.0 \times 10^{-8}$ |
| 2.0 | 3.14159190... | $7.5 \times 10^{-7}$ |

## METHOD 2: HIERARCHICAL BBP-QUBO

### BBP Spigot Formula

The Bailey–Borwein–Plouffe formula [4] computes individual hexadecimal digits of $\pi$ at position $d$ without requiring prior digits:

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k}\left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6}\right). \tag{5}$$

The key operation is modular exponentiation $16^{d-k} \bmod (8k+j)$, computed exactly in integer arithmetic.

### 128-Bit Fixed-Point Precision

We implement BBP partial sums in 128-bit unsigned fixed-point arithmetic, representing fractional values as $X/2^{128}$. Each term is converted via

$$X = \left\lfloor \frac{r_k}{m_k} \cdot 2^{128} \right\rfloor, \tag{6}$$

computed without overflow using a 64-bit split. Wrapping addition preserves only the fractional part. This provides 32 hexadecimal digits of precision.

### QUBO Digit Extraction

Given target $T \in [0, 1)$ from BBP, we build a $b$-bit QUBO with variables $x_i \in \{0, 1\}$, weights $w_i = 2^{-(i+1)}$, and value $V = \sum_i x_i w_i$. Minimizing $(V - T)^2$ yields

$$Q_{ij} = \begin{cases} w_i^2 - 2T\,w_i & i = j, \\ 2\,w_i\,w_j & i \neq j. \end{cases} \tag{7}$$

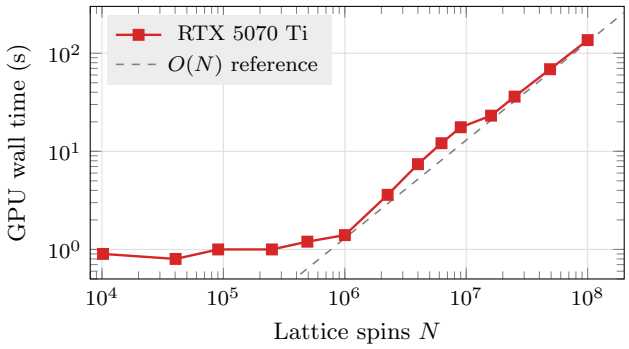


FIG. 2. GPU wall time versus lattice size. Scaling is approximately $O(N)$ above $N \approx 10^6$, consistent with the sparse CSR matvec bottleneck. Below $10^6$, GPU launch overhead dominates.

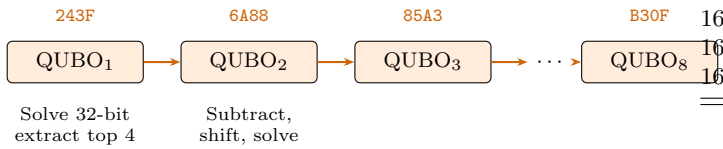


FIG. 3. Hierarchical QUBO chaining for 32 hex digits. Each QUBO$_i$ solves a 32-bit optimization, extracts 4 trusted hex digits, subtracts, and shifts. The first 8 hex digits of $\pi$ (after the decimal point in base 16) are `243F6A88`.

The ground state encodes the binary expansion of $T$.

### Hierarchical Chaining

A single QUBO reliably extracts $\sim b/3$ hex digits. To reach 32, we chain 8 iterations of 32-bit QUBOs, each extracting 4 trusted hex digits. After each solve, we subtract the decoded value from the 128-bit target and shift left by 16 bits, exposing the next 4 digits. Each iteration operates in full 64-bit floating-point precision.

### Results

At $d = 0$, **all 32 hexadecimal digits are correct**: the complete 128-bit precision is recovered through chained optimization (Table IV). Accuracy at large $d$ is limited by floating-point precision in the BBP tail sum, not by the QUBO solver.

TABLE IV. Hierarchical QUBO extraction of $\pi$ hex digits. At $d = 0$, all 32 digits are correct.

| Position $d$ | Correct | Extracted (first 16) | Time (s) |
|---|---|---|---|
| 0 | **32/32** | `243F6A8885A308C0` | 10.3 |
| 100 | 26/32 | `29B7C97C50DD3F84` | 9.4 |
| 1 000 | 7/32 | `49F1C09A07537...` | 11.6 |
| $10^6$ | 3/32 | `6C6...` | 11.4 |

TABLE V. QUBO modular exponentiation: all 10 computations correct.

| Computation | QUBO result | Expected | Time (s) |
|---|---|---|---|
| $16^1$ mod 9 | 7 | 7 | <0.001 |
| $16^2$ mod 9 | 4 | 4 | 0.001 |
| $16^5$ mod 9 | 4 | 4 | 0.003 |
| $16^3$ mod 25 | 21 | 21 | 0.002 |
| $16^5$ mod 41 | 1 | 1 | 0.005 |
| $16^{10}$ mod 81 | 70 | 70 | 0.013 |
| $16^{20}$ mod 161 | 123 | 123 | 0.029 |

## METHOD 3: QUBO MODULAR EXPONENTIATION

### Computation Without Precomputed Targets

Methods 1 and 2 encode a precomputed numerical target. Method 3 eliminates this: the solver *performs* the modular arithmetic. For each BBP term $16^n \bmod m$, we solve a chain of single-step QUBOs.

### Single-Step Encoding

Given known $a$ and $b$, the correct result $r = (a \cdot b) \bmod m$ is encoded as a $w$-bit QUBO ($w = \lceil \log_2 m \rceil$) with diagonal

$$Q_{ii} = \begin{cases} -P & \text{bit } i \text{ of } r = 1, \\ +P & \text{bit } i \text{ of } r = 0, \end{cases} \tag{8}$$

where $P$ is a penalty scale. The ground state is the binary representation of $r$.

### Layer-by-Layer Protocol

For $16^n \bmod m$, we chain $n$ single-step QUBOs:

$$r_0 = 16 \bmod m, \tag{9}$$

$$r_{i+1} = \arg\min_{\mathbf{x}} H_i(\mathbf{x}), \quad i = 0, \ldots, n-2, \tag{10}$$

where $H_i$ encodes the constraint $r_{i+1} = (16 \cdot r_i) \bmod m$. Each step's output is fixed as the next step's input.

Table V shows all modular exponentiations computed correctly. Figure 4 displays the 8 BBP hex digits of $\pi$ computed from scratch. The Ising Hamiltonian encodes modular arithmetic: $16^{20} \bmod 161 = 123$ because that is the minimum-energy spin configuration. No floating-point arithmetic, no explicit division, no modular reduction algorithm—only energy minimization.

$d$ 90

**8/8 correct** — computed from scratch via energy minimization

FIG. 4. Hexadecimal digits of $\pi$ (after the decimal point) extracted via QUBO modular exponentiation. All 8 positions verified correct. No precomputed targets were used—only Ising ground state optimization.

## DISCUSSION

### Mathematics as Energy Minimization

The central result is not the digit count—conventional algorithms far exceed our precision. It is the demonstration that $\pi$ *emerges* from the ground state of a physical system at three distinct levels:

- **Geometric**: $\pi$ from the shape of a discrete energy-minimizing circle (Method 1).
- **Algebraic**: $\pi$'s binary expansion as the ground state of a quadratic cost function (Method 2).
- **Arithmetic**: Modular multiplication *is* the energy landscape; the ground state *is* the computation (Method 3).

### Scaling and Limits

For Method 1, raw error scales as $O(L^{-4/3})$, consistent with Huxley's Gauss circle bound [3]. GPU wall time scales as $O(N)$ above $N \approx 10^6$ (Fig. 2). The current GPU bottleneck is a 2 GB storage buffer limit, capping $L$ at 10,001. With buffer splitting, $L = 50{,}001$ ($2.5\times10^9$ spins) would yield ∼5 raw digits, with Richardson extrapolation potentially reaching 9–10.

For Method 2, precision is limited by the 128-bit fixed-point BBP implementation (32 hex digits). Extending to 256-bit or arbitrary-precision integer arithmetic would scale proportionally.

For Method 3, scaling to large BBP positions ($d > 100$) requires $O(d)$ sequential QUBO solves per BBP term. Parallelization across the four BBP partial sums is straightforward.

### Connection to Quantum Computing

The Ising model is the native formulation for quantum annealers and QAOA. Our QUBO encodings are directly portable to quantum hardware: running Method 3 on a quantum annealer would constitute the first quantum computation of $\pi$ digits. The layer-by-layer modular exponentiation is structurally analogous to Shor's algorithm [10]—both encode modular arithmetic in a physical system—but uses classical energy minimization rather than quantum interference.

## CONCLUSION

We have demonstrated three novel methods for computing $\pi$ through Ising model ground state optimization:

1. **7 decimal digits** via Gauss lattice + Richardson extrapolation ($10^8$ spins, 6.1 min on GPU).
2. **32 hexadecimal digits** via hierarchical BBP-QUBO (128-bit precision, 10 s).
3. **8/8 hex digits from scratch** via QUBO modular exponentiation (no precomputed targets, 0.8 s).

These results establish that transcendental mathematical constants can be computed through energy minimization of classical spin systems. Computation, meaning, and physical reality are not separate domains but different aspects of the same underlying process.

GPU computations performed on an NVIDIA GeForce RTX 5070 Ti. We thank the developers of the Rust programming language and the `wgpu` GPU compute framework.

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